

Module 03 – Production Modeling

Model Formulation

Decision Variables: Units produced per month (484,404,545,57)

Objective Function: Total Cost (monthly production + monthly carrying costs)

Constraints: Units demanded, production capacity, minimum inventory

Model Optimized for Cost Reduction

	Month	1	2	3	4
Beginning Inventory		200	60	68	503
Units Produced		484	404	545	57
Units Demanded		624	396	110	509
Ending Inventory		60	68	503	51
Maximum Production		496	404	545	453
Minimum Inventory		60	40	11	51
Average Inventory		130	64	285.5	277
Unit Production Cost		\$ 48	\$ 48	\$ 56	\$ 46
Unit Carrying Cost		\$ 81	\$ 81	\$ 94	\$ 77
Monthly Production Cost		\$ 23,232	\$ 19,392	\$ 30,520	\$ 2,622
Monthly Carrying Cost		\$ 10,483	\$ 5,161	\$ 26,860	\$ 21,407
Total Cost	\$ 139,677				

My production and inventory model for the upcoming four months is designed to meet customer demand while minimizing total operational costs. By strategically adjusting production levels and carefully managing inventory, we aim to keep total costs at \$139,677 over the planning horizon.

Key Insights

- The model ensures production closely aligns with monthly demand while respecting production capacity limits.
- Production remains within monthly capacity limits, with peak utilization in Month 3 (545 units produced out of a 545-unit capacity).
- Ending inventories are maintained above minimum levels to avoid stockouts, ensuring smooth operations and customer satisfaction.